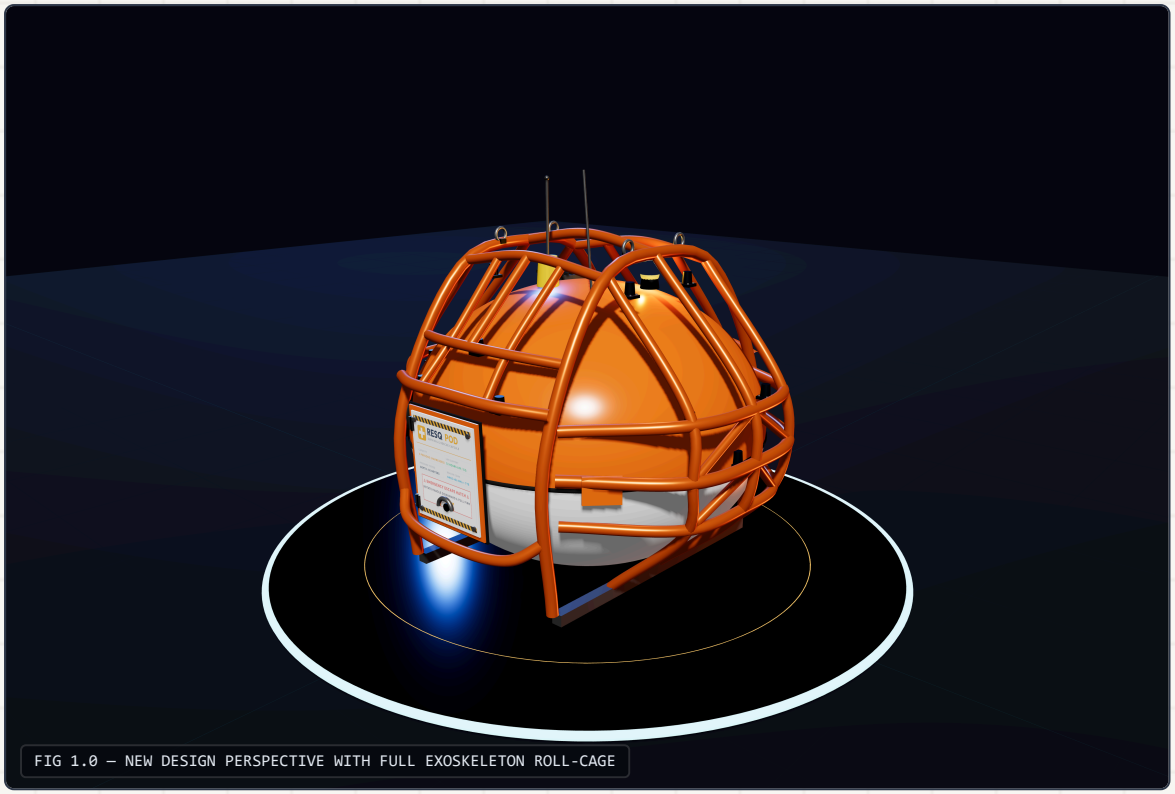

FIG 1.0 — NEW DESIGN PERSPECTIVE WITH FULL EXOSKELETON ROLL-CAGE

NEW DESIGN STRUCTURAL INNOVATIONS

PROTECTION UPGRADES

1. **Heavy-Duty Canopy Brow Bar**
   Overhead rigid deflector bar above hatch header shields door mechanism from falling rubble loads.
2. **Multi-Tier Side Crash Rub-Rails**
   Upper shoulder, mid equatorial, and lower deflector rails with diagonal space-frame web trusses.
3. **Full Rear Space-Frame Bumper**
   Twin vertical spine roll-bars, horizontal bumpers & X-braces for tumbling & rear impact protection.
4. **Upgraded Life Support Interior**
   Sub-floor bay, 3 racing seats, 72h O_2 cylinder, active CO_2 scrubber & keel acoustic pinger transponder.

MISSION PROFILE & CAPSULE CONCEPT

SYSTEM OVERVIEW

The **RESQ-POD** is a next-generation civil survival capsule engineered for catastrophic flooding, tsunamis, earthquakes, and structural building collapses. Designed to safeguard up to 3 occupants, the capsule delivers continuous life support for 24–72 hours while actively broadcasting multi-mode satellite and acoustic distress signals.

`3 Occupants (Harnessed)` `24–72h Closed Atmosphere` `120 kN Crush Tested` `+650 kg Reserve Buoyancy`

MASTER ENGINEERING SPECIFICATIONS

METRIC DATA

Overall Exterior Length	2,100 mm (2.10 m)
Overall Exterior Width (Hull)	1,600 mm (1.60 m)
Total Width with Side Cage Pipes	1,940 mm (1.94 m)
Overall Exterior Height	1,600 mm (1.60 m)
Internal Cabin Volume	2.85 m³
Dry Tare Mass (Empty)	285 kg
Fully Loaded Gross Mass (3 crew + cargo)	580 kg
Total Water Displacement	1,230 kg (+650 kg Buoyancy)
Hull Shell Material	Rotomolded HDPE (5–8 mm)
Internal Survival Frame	Tubular Ti / Al Alloy Skeleton
Exoskeleton Roll-Cage	Ø42mm 4130 Chromoly Steel
Impact Absorption Core	Expanded Polypropylene (EPP)
Flotation Core	Closed-Cell Hydrophobic Foam
Oxygen Storage	300 bar Carbon Composite (72h)
CO_2 Scrubbing Unit	Active Granular Soda-Lime + 12V
Distress Satellite Uplink	406 MHz Cospas-Sarsat + GNSS
Under-Rubble Locator	37.5 kHz Acoustic Pinger + TTE

RESQ-POD ORTHOGRAPHIC PROJECTIONS & GENERAL ARRANGEMENT

FRONT ELEVATION VIEW

WIDTH: 1,600 mm (1,940 mm CAGE) · HEIGHT: 1,600 mm

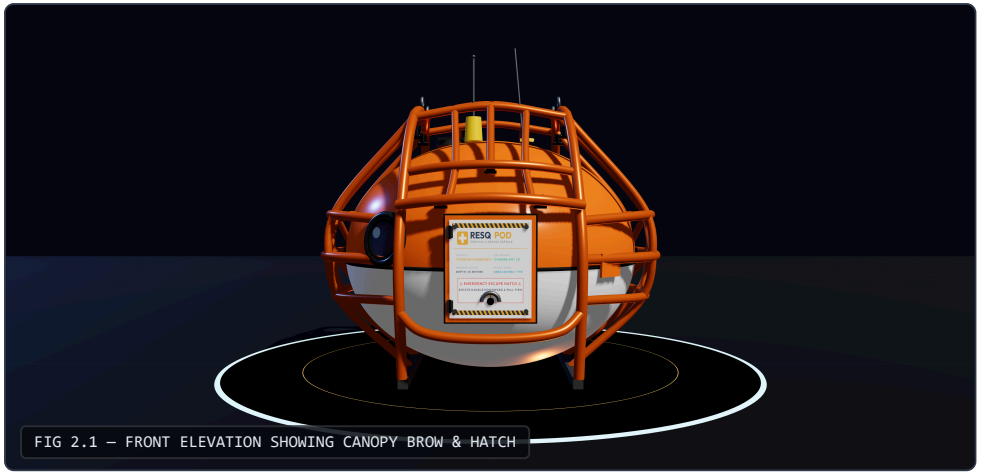

FIG 2.1 — FRONT ELEVATION SHOWING CANOPY BROW & HATCH

- **A** **Canopy Brow Bar:** Arched 42mm tubular roll bar directly above the hatch header protecting locking dog mechanism.
- **B** **Emergency Access Door:** Watertight gasketed hatch with dual inside/outside rotating lock handle.

SIDE ELEVATION PROFILE

LENGTH: 2,100 mm · SKID TRACK: 820 mm

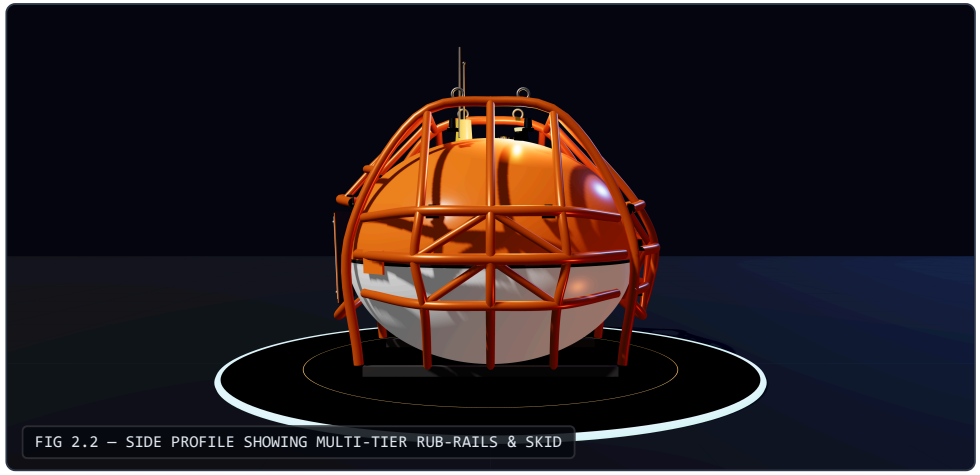

FIG 2.2 — SIDE PROFILE SHOWING MULTI-TIER RUB-RAILS & SKID

- **C** **Tri-Level Side Rub-Rails:** Upper shoulder rail, mid-equator rail, and lower deflector with triangulated web gussets.
- **D** **Ground Skids:** Dual longitudinal heavy tubular runners for ground stability and debris dragging.

TOP PLAN VIEW

ELLIPTICAL FOOTPRINT: 2,100 mm × 1,600 mm

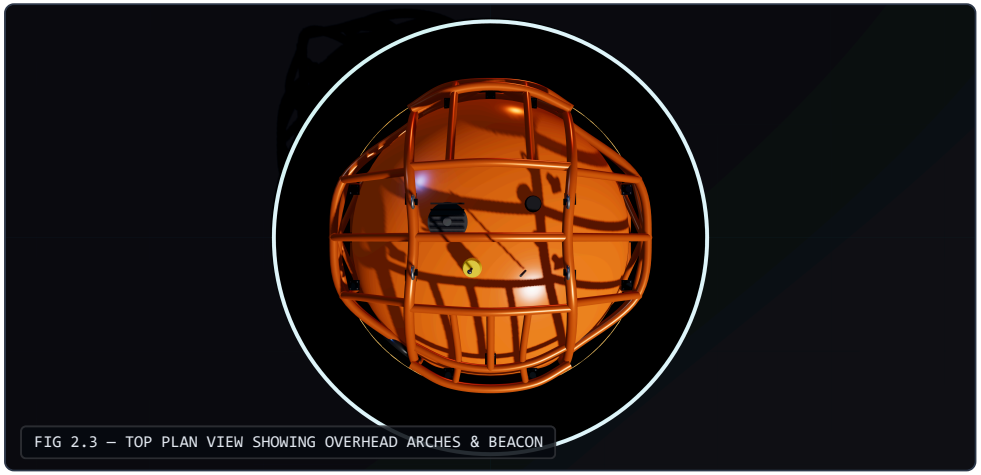

FIG 2.3 — TOP PLAN VIEW SHOWING OVERHEAD ARCHES & BEACON

- **E** **Overhead Roll Arches:** Twin longitudinal pipes protecting beacon, antennas, and passive relief vents.
- **F** **Distress Beacon Mast:** Pop-up 406 MHz antenna and high-intensity optical LED strobe beacon.

REAR ELEVATION VIEW

FULL REAR SPACE-FRAME BUMPER

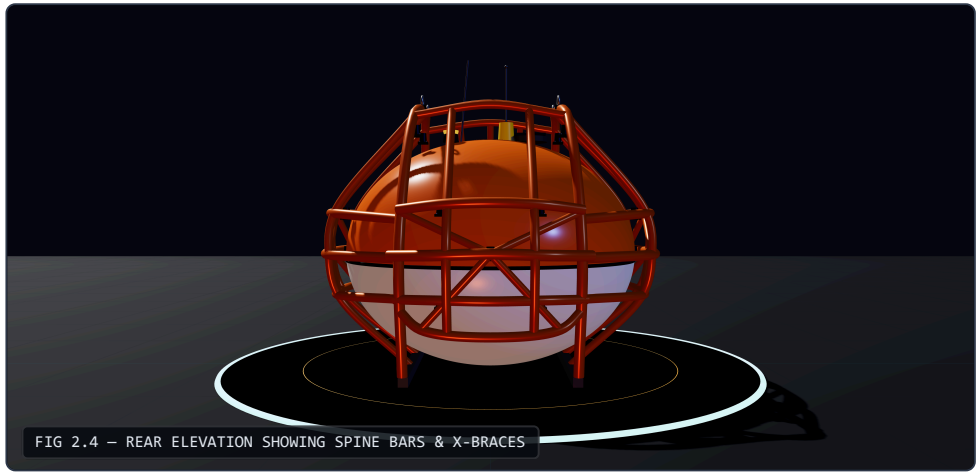

FIG 2.4 — REAR ELEVATION SHOWING SPINE BARS & X-BRACES

- **G** **Spine Roll-Bars & X-Brace:** Twin vertical spine members and diagonal X-braces absorbing rear impacts.
- **H** **Structural Hardpoint Standoffs:** Load-bearing standoffs transfer impact energy into internal frame

FIG 3.0 — TRANSVERSE CUTAWAY SECTION SHOWING COMPLETE 12-SYSTEM LAYOUT

DUAL-DECK ARCHITECTURE

CABIN BIOMECHANICS

The cabin incorporates a **Dual-Level Deck Configuration**. The upper seat deck positions crew at the neutral buoyancy center with 5-point harnesses for deceleration absorption. The sub-floor storage bay houses heavy life support essentials (water, rations, battery, and pinger), lowering the capsule center of mass (CoM) to guarantee automatic righting moment in water.

MASTER COMPONENT DIRECTORY (1–12)

SUBSYSTEM IDENTIFICATION

1. **Rotomolded HDPE Outer Shell:** Seamless 5–8mm high-visibility orange shell. Waterproof, corrosion-proof, UV-stabilized.
2. **Internal Structural Frame:** High-yield tubular titanium/aluminum skeleton maintaining interior survival space under heavy crushing loads.
3. **Closed-Cell Flotation Foam:** Symmetrically contoured hydrophobic polyurethane foam blocks ensuring positive buoyancy reserve (+650 kg).
4. **Thermal & Impact EPP Foam:** Multi-density Expanded Polypropylene lining absorbing decelerations and preventing hypothermia in freezing water.
5. **Compressed Oxygen Cylinder:** 300-bar carbon-composite cylinder providing 24–72 hours of oxygen metering for 3 adults.
6. **Active CO_2 Scrubber Unit:** Replaceable chemical soda-lime cartridge with brushless 12V fan keeping cabin CO_2 below 0.5%.
7. **Drinking Water Reserves:** 3x 4-liter heavy-duty sealed jerry cans (12 liters total drinking water) in the sub-floor bay.
8. **Emergency Food Rations:** High-calorie, vacuum-sealed emergency nutrition bars (10,000 kcal total, 5-year shelf life).
9. **GNSS + 406 MHz Emergency Beacon:** Cospas-Sarsat distress beacon with automated pop-up antenna and high-intensity strobe.
10. **Under-Rubble Acoustic Pinger:** Keel-mounted 37.5 kHz omnidirectional transponder enabling search teams to pinpoint buried pod.
11. **3x Impact-Attenuating Seats:** Ergonomic bucket seats with 5-point aviation-grade quick-release safety harnesses.
12. **Rugged Emergency Case & First Aid:** IP67 waterproof Pelican equipment case + Red Trauma First Aid Kit for onboard medical emergencies.

FRONT CANOPY BROW

HATCH PROTECTION

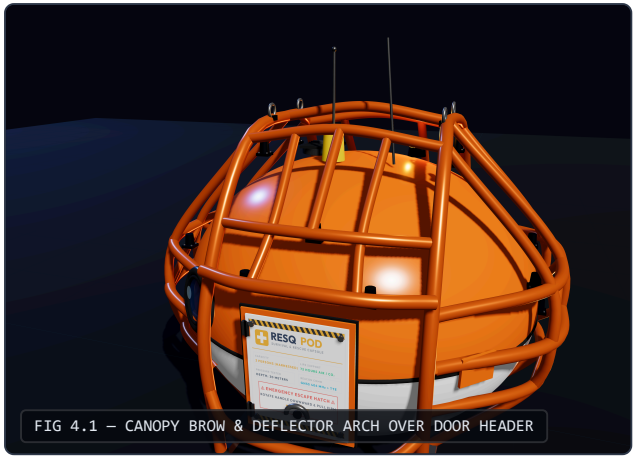

FIG 4.1 — CANOPY BROW & DEFLECTOR ARCH OVER DOOR HEADER

Debris Deflection Geometry:
In building collapse, debris frequently strikes door headers, deforming seals and trapping occupants. The Front Canopy Brow bridges from the overhead arch down to the door jambs with vertical gusset struts, creating a structural protective canopy.

Key Features:

- Prevents door frame pinch during roof crushing
- Shields multi-point locking dogs and hinges
- Provides solid grab handles for rescue divers

MULTI-TIER SIDE RUB-RAILS

LATERAL IMPACT

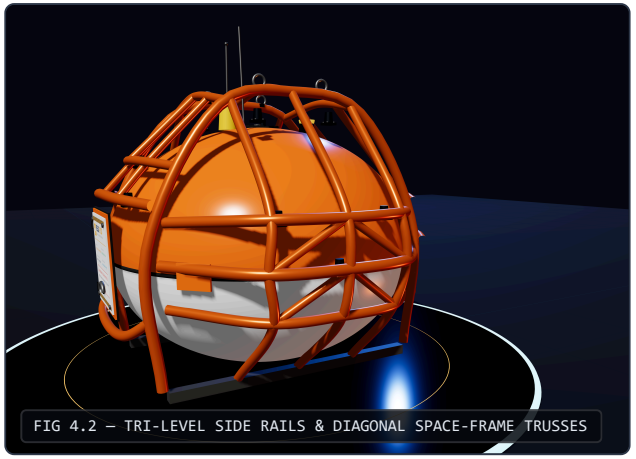

FIG 4.2 — TRI-LEVEL SIDE RAILS & DIAGONAL SPACE-FRAME TRUSSES

Space-Frame Triangulation:
Lateral impacts against concrete piers or floating debris are resisted by a three-tier horizontal cage (Upper Shoulder, Mid Equator, Lower Deflector) joined by diagonal web gussets.

Load Distribution:

- 18 hardpoint standoff mounts isolate the HDPE hull
- Deflects puncturing objects away from porthole
- Full energy absorption prior to hull elastic limit

REAR FRAME & SKID BASE

TUMBLE DEFENSE

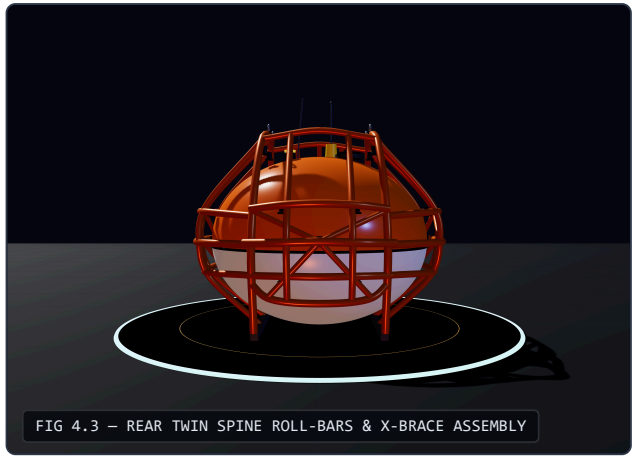

FIG 4.3 — REAR TWIN SPINE ROLL-BARS & X-BRACE ASSEMBLY

Material Alloy	4130 Seamless Steel
Outer Diameter (OD)	Ø 42.0 mm
Wall Thickness	3.2 mm (11 ga)
Yield Strength	435 MPa (63 ksi)
Tensile Strength	670 MPa (97 ksi)
Weld Specification	GTAW / TIG Full Pen
Finish / Coating	Polyester Powdercoat
Corrosion Standard	1,000h Salt Spray
Skid Wear Runners	UHMW-PE Replaceable

5-LAYER COMPOSITE HULL STRATIGRAPHY

CROSS-SECTION MATRIX

1. **Rotomolded HDPE Outer Shell (5–8 mm):** High-density polyethylene, waterproof, impact & puncture resistant, corrosion-proof.
2. **Expanded Polypropylene EPP Foam (50–100 mm):** Progressive kinetic energy absorption foam, dampening external shocks.
3. **Titanium / Aluminium Structural Skeleton:** Rigid tubular frame preventing hull buckling under extreme crushing pressures.
4. **Closed-Cell Flotation Foam:** Zero-water-ingress polyurethane barrier delivering permanent positive reserve buoyancy.
5. **Washable Antimicrobial Fire-Resistant Liner:** FAR 25.853 compliant soft-touch interior padding and acoustic insulation.

Hull Survivability Standards

The 5-layer composite structure provides thermal insulation against water temperatures from -2°C to 45°C. The exterior HDPE prevents water ingress, while the EPP and closed-cell layers ensure that even if the outer shell suffers puncture damage, the capsule remains buoyant and thermally stable.

WATERTIGHT HATCH

Dual automotive-grade EPDM gaskets. Multi-point rotary deadbolts with dual inside/outside release levers for emergency egress.

SATELLITE UPLINK

406 MHz Cospas-Sarsat transmitter automatically sends GPS distress signal on water release. High-visibility optical LED strobe beacon.

LIFE SUPPORT / O_2

300-bar carbon-composite cylinder supplies 0.5 L/min per occupant. Soda-lime scrubber scrubs exhaled CO_2 below 5,000 ppm.

ACOUSTIC PINGER

Keel-mounted 37.5 kHz omnidirectional transponder penetrates up to 15m of solid concrete, water, or mud for 30 days.

1 Surface / Flood Inundation

- Low Center of Mass (CoM) ensures self-righting stability in violent surf and torrents.
- Closed-cell foam ensures unsinkability even if hull is breached.
- Pop-up GNSS beacon establishes immediate satellite distress fix.
- High-visibility International Safety Orange livery aids aerial search.

2 Structural Debris Burial

- Full 4130 exoskeleton roll-cage resists 120 kN distributed crushing rubble loads.
- Canopy brow bar protects door header, preventing door pinch.
- Keel acoustic pinger pulses 37.5 kHz signals detectable through deep rubble.
- 72h onboard O_2 & scrubber sustain life while search teams excavate.

3 First-Responder Extrication

- Top-mounted heavy-duty forged crane lifting eyes for helicopter or winch extraction.
- External emergency door release allows rescuers rapid entry.
- Sub-floor medical kit immediately accessible to treat injured crew.
- Standardized skid base fits standard flatbed truck and helicopter skids.